

BIOL-8 Exam 03

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: _____

Date: _____

Modules 11-15: Skeletal, Muscular, Nervous, Microbiology & Cardiopulmonary Systems

Date: Thursday, April 16, 2026

Time: 5:30 PM - 8:40 PM

Total Points: 50

Part A: Multiple Choice (31 points)

Select the best answer for each question. Each question is worth 1 point.

Module 11: Skeletal System

1. Which of the following is **NOT** a function of the skeletal system?
 - A) Hormone secretion for digestion
 - B) Support
 - C) Protection of internal organs
 - D) Blood cell production

2. Which statement best describes how the skeletal and muscular systems work together to produce body movement?

- A) Bones contract and pull on muscles across joints
- B) Skeletal muscles contract and pull on bones across joints; bones act as levers
- C) Muscles push bones directly without using joints
- D) Only smooth muscle attaches to bones for voluntary movement

3. Where does hematopoiesis (blood cell formation) occur?

- A) Compact bone only
- B) Yellow bone marrow only
- C) Red bone marrow
- D) Periosteum

4. The epiphyseal (growth) plate is made of:

- A) Compact bone
- B) Spongy bone only
- C) Fibrous connective tissue only
- D) Cartilage

5. The axial skeleton includes all of the following EXCEPT:

- A) Femur
- B) Skull
- C) Vertebral column
- D) Rib cage

6. A freely movable joint (like the knee or shoulder) is classified as:

- A) Fibrous
- B) Synovial
- C) Cartilaginous
- D) Sutured

Module 12: Muscular System

1. Which of the following is a function of skeletal muscle?
 - A) Pumping blood (cardiac muscle)
 - B) Lining the walls of blood vessels only
 - C) Moving the skeleton and generating body heat
 - D) Controlling pupil size only
2. The correct order from largest to smallest in skeletal muscle organization is:
 - A) Myofibril → Fiber → Fascicle → Muscle
 - B) Fiber → Muscle → Fascicle → Myofibril
 - C) Sarcomere → Muscle → Fascicle → Fiber
 - D) Muscle → Fascicle → Fiber (cell) → Myofibril
3. The basic contractile unit of a muscle fiber is the:
 - A) Sarcomere
 - B) Fascicle
 - C) Myofibril
 - D) Motor unit
4. The thick filaments in a sarcomere are made of:
 - A) Actin
 - B) Myosin
 - C) Troponin
 - D) Tropomyosin
5. According to the sliding filament theory, during contraction:
 - A) The thick and thin filaments get shorter
 - B) The sarcomere stretches apart
 - C) The thin filaments slide past the thick filaments, shortening the sarcomere
 - D) New filaments are synthesized

6. The neurotransmitter released at the neuromuscular junction is:

- A) Dopamine
- B) Serotonin
- C) Epinephrine
- D) Acetylcholine (ACh)

Module 13: Nervous System

1. The central nervous system (CNS) consists of:

- A) The brain and spinal cord
- B) All nerves outside the brain
- C) The sensory organs only
- D) The autonomic nerves only

2. The peripheral nervous system (PNS) includes:

- A) The brain and spinal cord
- B) All nerves and ganglia outside the brain and spinal cord
- C) Only motor nerves
- D) Only the spinal cord

3. The division of the PNS that controls voluntary movements of skeletal muscles is the:

- A) Autonomic nervous system
- B) Parasympathetic division
- C) Somatic nervous system
- D) Sympathetic division

4. The long extension of a neuron that carries signals away from the cell body is the:

- A) Dendrite
- B) Soma
- C) Myelin sheath
- D) Axon

5. What is the function of myelin?

- A) It insulates the axon and speeds up signal transmission
- B) It produces neurotransmitters
- C) It stores calcium ions in the soma
- D) It connects neurons to glands only

6. A reflex arc differs from conscious sensation because:

- A) It does not require a stimulus
- B) It can be processed at the spinal cord level so the initial response does not require the brain
- C) It only involves the cerebellum
- D) It is always voluntary

Module 14: Microbiology

1. A typical bacterial cell is different from a plant or animal cell mainly because it:

- A) Has no DNA
- B) Has no cytoplasm
- C) Has no membrane around its nucleus (its DNA is in the nucleoid region)
- D) Is always larger than a human cell

2. The three common shapes of bacteria are:

- A) Round, flat, and square
- B) Gram-positive, Gram-negative, and neutral
- C) Aerobic only, anaerobic only, and mixed
- D) Cocci (spheres), bacilli (rods), and spirilla (spirals)

3. In the Gram stain, Gram-positive bacteria usually look purple because they:

- A) Have a thick cell wall and no outer membrane
- B) Have no cell wall
- C) Have only a thin wall plus an outer membrane (typical of many Gram-negative bacteria)

- D) Are always harmless

4. One bacterium splits into two mostly by:

- A) Mitosis
- B) Binary fission
- C) Meiosis
- D) Fertilization

5. After people use antibiotics, resistant bacteria can become more common because:

- A) The drug trains bacteria to "try harder"
- B) The immune system stops working
- C) Bacteria that survive can reproduce and pass resistance to the next generation
- D) All bacteria die at the same rate

6. Many biologists say a virus is not a living cell because a virus:

- A) Is too small to see
- B) Never has genetic material
- C) Only lives in soil
- D) Cannot make its own energy or copy itself without infecting a host cell

Module 15: Cardiopulmonary System

1. Blood coming back from the body (low in oxygen) enters the heart first in the:

- A) Right atrium
- B) Left atrium
- C) Left ventricle
- D) Aorta

2. Which statement about **oxygen (O₂)** and **carbon dioxide (CO₂)** in the lungs is **true**?

- A) Oxygen moves from the blood into the air in the alveoli
- B) Carbon dioxide moves from the blood into the alveolar air so you can exhale it
- C) You breathe in carbon dioxide from the room air to feed the blood

- D) O₂ and CO₂ never cross between the air and the blood

3. Which type of vessel carries blood **away from** the heart?

- A) Veins
- B) Capillaries
- C) Arteries
- D) Lymph vessels

4. The first heart sound ("lub") is made when the valves **between the upper and lower heart chambers** close. Those valves are the:

- A) Semilunar valves (at the exits of the ventricles to the aorta and pulmonary artery)
- B) Valves in the leg veins
- C) Lung valves only
- D) AV (atrioventricular) valves

5. In the lungs, oxygen moves into the blood mainly in the tiny air sacs called the:

- A) Alveoli
- B) Trachea
- C) Bronchi
- D) Larynx

6. Most of the oxygen in your blood is:

- A) Dissolved in the plasma
- B) Carried by hemoglobin in red blood cells
- C) Stored in white blood cells
- D) Attached to platelets

7. When you breathe in quietly, the diaphragm contracts and moves:

- A) Up, so the chest gets smaller
- B) Sideways only
- C) Down, so the chest gets larger

- D) Only when you breathe out

Part B: Fill in the Blank (10 points)

Complete each statement with the correct term from the word bank. Each blank is worth 1 point.

Word Bank: *acetylcholine, actin, appendicular, axial, capsid, dendrites, peptidoglycan, periosteum, sympathetic, troponin*

1. The tough membrane covering the outer surface of bone is the _____.
 2. The skull, vertebral column, and rib cage belong to the _____ skeleton.
 3. The limbs and girdles belong to the _____ skeleton.
 4. The thin filaments of the sarcomere are made mainly of the protein _____.
 5. Calcium binds to _____, moving tropomyosin so that myosin can bind to actin.
 6. The autonomic division that dominates during the "fight-or-flight" response is the _____ division.
 7. The branched structures that receive incoming signals from other cells are _____.
 8. The cell wall component that is especially important when comparing Gram-positive and Gram-negative bacteria is _____.
 9. A typical virus has a nucleic acid genome surrounded by a protein coat called the _____.
 10. At the neuromuscular junction, the motor neuron releases _____ to start muscle contraction.
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Part C: Free Response (9 points)

Choose *THREE* of the following five questions. Answer in the provided spaces below. You may write, draw, or both. Each response is worth 3 points.

1. **Bone cells.** Compare osteoblasts, osteocytes, and osteoclasts. What does each cell type do, and why are all three important for bone health?
2. **ATP and calcium.** Explain why both ATP and calcium are required for skeletal muscle contraction. What happens if either is unavailable?
3. **Autonomic divisions.** Explain the difference between the sympathetic and parasympathetic divisions. Give one example of how each affects an organ (for example, heart rate or digestion).
4. **Antibiotic resistance.** Explain in your own words why antibiotic resistance is an example of natural selection. Use the terms *variation*, *selection*, and *reproduction* in your answer.
5. **Cardiopulmonary integration.** Trace the path of a drop of blood from the right atrium through the lungs and back to the left side of the heart. Then name one way the cardiovascular and respiratory systems work together during exercise.

Response 1

Question Number: _____

Response 2

Question Number: _____

Response 3

Question Number: _____

Feedback (Optional)

A. Is there anything else you would like to share (e.g., questions, comments, or notes for the professor)?

B. What percentage grade do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

End of Exam 03